

2023-Ph-H1-Q8

Section: Electricity

Topic: Monitoring and Measuring a.c.

An oscilloscope shows a sine wave with a peak voltage of 10 V. What is the r.m.s. voltage?

Answer: B

(7.1 V)

Guidance for Students:

The r.m.s. value for a sine wave is $V_{\text{rms}} = \frac{V_{\text{peak}}}{\sqrt{2}}$. For 10 V peak:
 $V_{\text{rms}} = \frac{10}{\sqrt{2}} = 7.07 \text{ V} \approx 7.1 \text{ V}$

Revision Tips:

- $V_{\text{rms}} = 0.707 \times V_{\text{peak}}$
- r.m.s. is like the d.c. equivalent value.
- Always use r.m.s. in power calculations.